

EMG4 MLM analyses

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Overview

This document contains the statistical analyses of the EMG4 data.

Building the models

Because our main focus was on the development of the corrugator response over time, we used a growth curve model approach (Mirman, 2015; Peck and Devore, 2008) with specific analysis designs that were optimized for assessing and comparing time trends across conditions. We first modelled participants and items as random factors. To assess the effect of our manipulations on the average activation across an entire segment, we first added morality dimension (moral vs. immoral) as a fixed factor in the model for the character manipulation segment, and morality dimension (moral vs. immoral), critical event valence (positive vs. negative) and interaction as fixed factors in the model for the affective state adjective and affect reason segments. Afterwards the interaction was added to the random part of the model as a random slope.

Next, linear, quadratic, and cubic trends will be added as covariates in the fixed part of the model, and if significantly improving the model also to the random part, see below. Time trend (e.g. linear) components will be added per condition to maintain flexibility in building the model, and to avoid forcing the model to fit, for example, a linear trend for all conditions when only one condition contains a significant linear component. All trend components will be centered to avoid correlation between trends (fixed effects final model intercepts therefore reflect the average corrugator activity across the entire segment, not the level at which corrugator activity intercepts the y-axis). By using trends up to the cubic component, we achieve

some flexibility to fit responses without over-fitting or losing explanatory power (Mirman, 2015). Because we are particularly interested in temporal developments, we will add random slopes for subjects for each time trend that initially improves the model (as well as standard random intercepts for subject and item).

1. Morality segment

variables:

dependent variable: Morality_response random factors: sjCode, Markertrial

independent variables: Morality (102=moral, 103=immoral), StateAdjectiveCondition (104=positive, 105=negative), t1_ (continuous linear trend per condition, will be centered for easier interpretation of the estimates), t2_ (continuous quadratic trend per condition, will be centered for easier interpretation of the estimates), t3_ (continuous cubic trend per condition, will be centered for easier interpretation of the estimates)

Summary of best Model MoralityM12

The best model is Model 12: Morality_response ~ MoralityCondition + t1_immoral + t1_moral + t2_immoral + t3_immoral + (1 + MoralityCondition | sjCode) + (1 | Markertrial). This is the summary of that model.

```
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_Morality_final_model_60sjs.RData"))
```

```
tab_model(MoralityM12)
```

Character Morality Segment (M12, df= 186989)			
Predictors	Estimates (95% CI)	z	p
(Intercept)	111.54 (104.87, 118.20)	32.82	<0.001*
MoralityCondition [Immoral]	49.51 (26.11, 72.91)	4.15	<0.001*
t1 immoral	12.25 (10.68, 13.83)	15.25	<0.001*
t1 moral	-1.66 (-2.29, -1.03)	-5.18	<0.001*
t2 immoral	-5.00 (-5.48, -4.51)	-20.08	<0.001*
t3 immoral	-0.51 (-0.89, -0.12)	-2.59	0.01*
Random Effects			
σ^2	20072.72		
τ_{00} Markertrial	251.52		
τ_{00} sjCode	444.25		
τ_{11} sjCode.MoralityConditionImmoral	8509.90		
ρ_{01} sjCode	0.47		
ICC	0.23		
Deviance	2384129.40		
log-Likelihood	-1192064.70		

Post-hoc analyses

```
MoralityM12_emmeans_contrasts = emmeans(MoralityM12, specs = tukey ~ MoralityCondition, lmer.
df = "satterthwaite", lmerTest.limit=187000, simple = "each", combine = T, adjust = "sidak",
reverse = T);
MoralityM12_emmeans_contrasts$emmeans
```

```
## MoralityCondition emmean      SE      df lower.CL upper.CL
## Moral              106    3.41  105.7     99.6     113
```

```
## Immoral 156 13.55 62.6 128.8 183
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95

MoralityM12_emmeans_contrasts$contrasts %>%
  summary(infer = TRUE)

## contrast estimate SE df lower.CL upper.CL t.ratio p.value
## Immoral - Moral 49.5 11.9 60 25.6 73.4 4.147 0.0001
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

Contrasts are redundant, as this is already given in the summary of the model. Also the comparison between t1_immoral and t1_moral is redundant since the estimates are tested against 0 and their signs differ, therefore the difference between t1_immoral and t1_moral is also significantly different.

2. State Adjective segment

variables:

dependent variable: StateAdjective_response random factors: sjCode, Markertrial

independent variables: Morality (102=moral, 103=immoral), StateAdjectiveCondition (104=positive, 105=negative), t1_ (continuous linear trend per condition, will be centered for easier interpretation of the estimates), t2_ (continuous quadratic trend per condition, will be centered for easier interpretation of the estimates), t3_ (continuous cubic trend per condition, will be centered for easier interpretation of the estimates)

Summary of best Model AdjectiveM17

The best model is Model 17: StateAdjective_response ~ MoralityCondition + StateAdjectiveCondition + MoralityCondition:StateAdjectiveCondition + t1_moral_neg + t1_moral_pos + t2_immoral_neg + t2_moral_neg + (1 + t1_moral_neg + t2_immoral_neg | sjCode) + (1 | Markertrial). This is the summary of that model.

```
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_StateAdjective_final_model_60sjs.RData"))
tab_model(AdjectiveM17)
```

Predictors	State Adjective Segment (M17, df = 56084)		
	Estimates (95% CI)	z	p
(Intercept)	110.74 (99.48, 122.00)	19.27	<0.001*
MoralityCondition [Immoral]	6.48 (4.37, 8.60)	6.00	<0.001*
StateAdjectiveCondition [Negative]	22.68 (19.97, 25.38)	16.44	<0.001*
t1 moral negative	8.25 (-3.41, 19.92)	1.39	0.17
t1 moral positive	-5.47 (-8.94, -2.00)	-3.09	0.002*
t2 immoral negative	-11.67 (-35.01, 11.68)	-0.98	0.33
t2 moral negative	-34.89	-7.60	<0.001*

	(-43.89, -25.89)		
MoralityCondition			
[Immoral] *			
StateAdjectiveCondition			
[Negative]	-11.36	-5.83	<0.001*
	(-15.18, -7.54)		
Random Effects			
σ²	8159.42		
τ₀₀ Markertrial	112.07		
τ₀₀ sjCode	1840.47		
τ₁₁ sjCode.t1_moral_negative	1936.11		
τ₁₁ sjCode.t2_immoral_negative	7239.30		
ρ₀₁ sjCode.t1_moral_negative	0.16		
ρ₀₁ sjCode.t2_immoral_negative	-0.33		
ICC	0.20		
Deviance	665242.55		
log-Likelihood	-332621.28		

Post-hoc analyses

```

AdjectiveM17_emmeans_contrasts = emmeans(AdjectiveM17, specs = tukey ~ MoralityCondition * StateAdjectiveCondition, lmer.df = "satterthwaite", lmerTest.limit=56100, simple = "each", combine = T, adjust = "sidak", reverse = T);
AdjectiveM17_emmeans_contrasts$emmeans

## MoralityCondition StateAdjectiveCondition emmean SE df lower.CL upper.CL
## Moral Positive 109 5.61 69.2 97.4 120
## Immoral Positive 115 5.61 69.2 103.9 126
## Moral Negative 131 5.64 70.8 120.0 142
## Immoral Negative 126 5.64 70.8 115.1 138
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95

AdjectiveM17_emmeans_contrasts$contrasts %>%
  summary(infer = TRUE)

## StateAdjectiveCondition MoralityCondition contrast estimate SE
## Positive . Immoral - Moral 6.48 1.08
## Negative . Immoral - Moral -4.88 1.62
## . Moral Negative - Positive 22.68 1.38
## . Immoral Negative - Positive 11.31 1.38
## df lower.CL upper.CL t.ratio p.value
## 55857 3.79 9.177 5.998 <.0001
## 55849 -8.92 -0.837 -3.007 0.0105
## 55863 19.24 26.112 16.440 <.0001
## 55861 7.88 14.748 8.209 <.0001
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: sidak method for 4 tests

```

Plot of regression lines for the different State Adjectives for the Morality manipulation in the State Adjective segment:



Figure on State Adjective segment with CI

The comparison between t1_moral_neg and t1_moral_pos is redundant since the estimates are tested against 0 and their signs differ, therefore the difference between t1_moral_neg and t1_moral_pos is also significantly different.

3. Affect Reason segment

variables:

dependent variable: AffectReason_response random factors: sjCode, Markertrial

independent variables: Morality (102=moral, 103=immoral), StateAdjectiveCondition (104=positive, 105=negative), t1_ (continuous linear trend per condition, will be centered for easier interpretation of the estimates), t2_ (continuous quadratic trend per condition, will be centered for easier interpretation of the estimates), t3_ (continuous cubic trend per condition, will be centered for easier interpretation of the estimates)

Summary of best Model ReasonM22

The best model is Model 22: AffectReason_response ~ MoralityCondition + StateAdjectiveCondition + MoralityCondition:StateAdjectiveCondition + t1_moral_neg + t1_moral_pos + t2_moral_neg + t3_moral_neg + (1 + t1_moral_neg + t3_moral_neg | sjCode) + (1 | Markertrial). This is the summary of that model.

```
load(paste0(getwd(), "/WORCS_EMG4/results/EMG4_AffectReason_final_model_60sjs.RData"))
tab_model(ReasonM22)
```

Predictors	Affect Reason Segment (M22, df = 93484)		
	Estimates (95% CI)	z	p
(Intercept)	104.72 (94.63, 114.82)	20.33	<0.001*

MoralityCondition [Immoral]	13.56 (11.64, 15.48)	13.84	<0.001*
StateAdjectiveCondition [Negative]	38.99 (36.54, 41.43)	31.24	<0.001*
t1 moral negative	6.80 (-0.28, 13.89)	1.88	0.06
t1 moral positive	-3.82 (-5.71, -1.94)	-3.98	<0.001*
t2 immoral negative	-14.56 (-17.47, -11.64)	-9.78	<0.001*
t3 moral negative	-5.03 (-11.31, 1.24)	-1.57	0.12
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	-35.14 (-38.25, -32.03)	-22.17	<0.001*
Random Effects			
σ^2	11172.06		
τ_{00} Markertrial	154.63		
τ_{00} sjCode	1417.61		
τ_{11} sjCode.t1_moral_negative	434.86		
τ_{11} sjCode.t3_moral_negative	278.58		
ρ_{01} sjCode.t1_moral_negative	0.56		
ρ_{01} sjCode.t3_moral_negative	-0.76		
ICC	0.12		
Deviance	1137450.49		
log-Likelihood	-568725.25		

Post-hoc analyses

```
ReasonM22_emmeans_contrasts = emmeans(ReasonM22, specs = tukey ~ MoralityCondition * StateAdjectiveCondition, lmer.df = "satterthwaite", lmerTest.limit=93500, simple = "each", combine = T, adjust = "sidak", reverse = T)
ReasonM22_emmeans_contrasts$emmeans
```

```
## MoralityCondition StateAdjectiveCondition emmean SE df lower.CL upper.CL
## Moral Positive 103 5.15 74.1 92.6 113
## Immoral Positive 116 5.15 74.1 106.1 127
## Moral Negative 142 5.18 75.7 131.5 152
## Immoral Negative 120 5.15 74.1 110.0 131
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

```
ReasonM22_emmeans_contrasts$contrasts %>%
  summary(infer = TRUE)
```

```
## StateAdjectiveCondition MoralityCondition contrast estimate SE
## Positive . Immoral - Moral 13.56 0.980
## Negative . Immoral - Moral -21.58 1.246
## . Moral Negative - Positive 38.99 1.248
## . Immoral Negative - Positive 3.85 0.978
## df lower.CL upper.CL t.ratio p.value
## 93266 11.12 16.00 13.839 <.0001
## 93261 -24.68 -18.48 -17.322 <.0001
## 93270 35.88 42.10 31.240 <.0001
## 93274 1.41 6.29 3.936 0.0003
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

```
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: sidak method for 4 tests
```

Plot of regression lines for the different State Adjectives for the Morality manipulation in the Affect Reason segment:

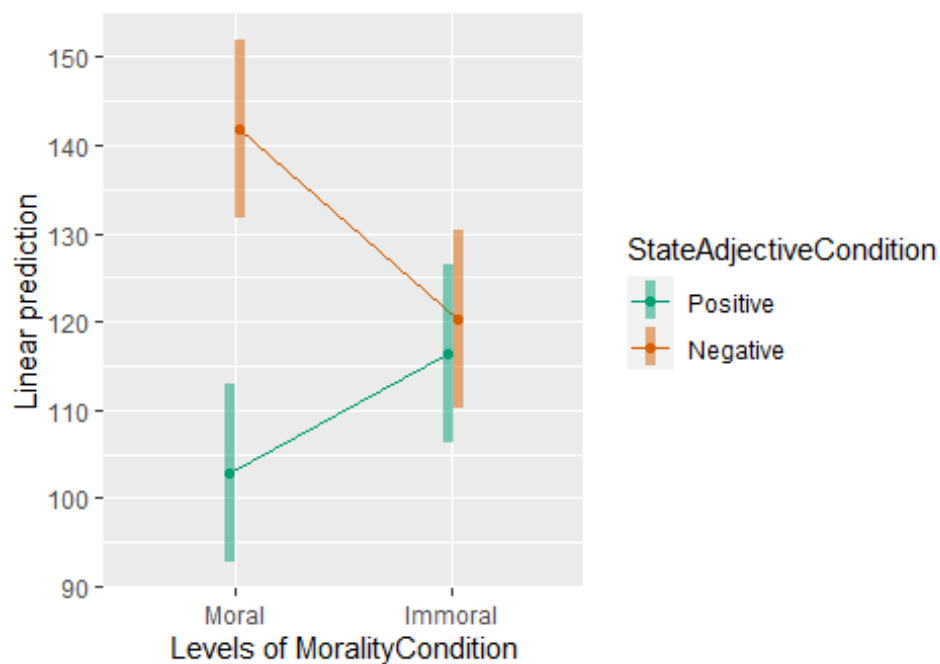


Figure on Affect Reason segment with CI

The comparison between t1_moral_neg and t1_moral_pos is redundant since the estimates are tested against 0 and their signs differ, therefore the difference between t1_moral_neg and t1_moral_pos is also significantly different.